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All Aboard? Language, Multimodality and Epistemic Inclusion in Seychelles Primary 4 Examinations

Mats Deutschmann

1. Introduction

Exam-oriented education systems remain a central and defining feature of schooling across many former British colonies in sub-Saharan Africa (SSA). Seychelles is no exception. These educational models are part of a powerful colonial legacy that positions centrally administered examinations, often based on British prototypes, as the primary instruments for defining curriculum content and for practical aspects such as ranking, selection, and certification. In particular, national examinations carry high stakes: they regulate the grouping and streaming of students across successive levels of education, thereby determining individual learners' academic trajectories and future opportunities. They also shape the perceived quality and status of schools, thereby indirectly influencing resource and competence allocation. The real and symbolic authority of exams within the system thus makes them a key control device through which educational priorities are articulated and enforced.

This centrally steered exam focus has far-reaching consequences. A substantial body of research (see, for example, Bartlett and Vavrus, 2013; D'Agostino, 2023; Jiang, 2024; Joong and Gibbs, 2025) highlights how the dominance of high-stakes national examinations can negatively affect teaching and learning in the field, narrowing the curriculum and privileging rote memorization and exam performance over the more creative and critical application of knowledge. In these exam-oriented contexts, classroom practices are often tightly aligned with examination demands, to the extent that, as Allen et al. (2016, p.ii) observe: 'little priority is given in classrooms to matters that are not closely aligned to the demands of examinations.'

In Seychelles, where Kreol Seselwa is the dominant spoken first language (85.1% of the population according to the Seychelles Population and Housing Census (National Bureau of Statistics, 2022)), national examinations are conducted almost exclusively in English. English holds official status within the education system, and assessments rarely account for the linguistic realities of students. Nor do they provide meaningful adaptations for second-language users, especially important during the early years after transition to English medium of instruction and assessment (hereafter MoIA) in Primary 3. As a result, examinations that are intended to measure subject knowledge and cognitive ability often end up reflecting learners' proficiency in English rather than their actual understanding of

content (cf. Bartlett and Vavrus, 2013). As Clegg (2007, p.43) observes: 'If we assess children in a second language it may not tell us what they know.'

This disconnect is further exasperated by the limited incorporation of local cultural and linguistic references in exam content, something which can further disadvantage learners whose everyday experiences and ways of knowing are not reflected in assessment tasks. This is especially true when exam questions are based on prototypical question formats originally produced for British contexts. Consequently, dis/connect to local cultural, as well as linguistic reference points, becomes a significant gatekeeping mechanism, shaping who is able to demonstrate knowledge successfully. These dynamics raise important concerns about fairness and validity, as well as the broader implications of an exam-driven system in which performance in English and familiarity with Western cultural frameworks shape opportunities for educational progression. Such demands thus play a critical role in processes of epistemic inclusion and exclusion (c.f. Barrett, 2025).

Moreover, these challenges intersect with socio-economic and gender inequalities. Learners from lower socio-economic backgrounds often have more limited exposure to English and Western media outside the classroom, particularly in homes where Kreol Seselwa predominates, placing them at a systemic disadvantage (Hungi and Thuku, 2010). Gendered patterns also emerge, with boys frequently underperforming relative to girls, a disparity often linked to comparatively weaker language proficiency and engagement with the linguistic demands of schooling (Hungi and Thuku, 2010; Deutschmann and Zelime, 2015). Together, these factors highlight how language in examinations is not merely a technical issue, but a broader social justice concern with significant implications for equity and social mobility in the Seychelles context.

Research investigating the nature and impact of high-stakes exams in Seychelles is limited. There have been some attempts to identify language support needs in examination contexts (Deutschmann and Zelime, 2025) as well as a few intervention studies that explicitly explore the impact of linguistic aspects on outcomes in assessment tasks (Zelime et al., 2018; Zelime and Deutschmann, 2019). To my knowledge, however, this study constitutes a first attempt to specifically analyse linguistic and cultural aspects of national examination content in relation to student outcomes.

More specifically, this study explores linguistic and cultural aspects that may have an impact on epistemic inclusion in the Primary 4 national exams, the first high-stakes national exams taken after transition to English MoIA in Seychelles. The aim is to investigate whether and how language use and cultural factors may hinder or support Seychellois learners in understanding exam questions, thereby preventing or enabling them to demonstrate their knowledge. Focus is on two curriculum subjects, English language and science.

The article consists of seven sections. Next, in Section 2, I set out the aim and research questions in more detail. This is followed by a contextual overview in Section 3, where I

give a brief description of the current national examination system and an overview of the language-in-education policies in relation to the concept of epistemic inclusion. In Section 4, I provide a brief review of previous research on assessment in language transition education systems in the region. Methods and materials are described in Section 5, followed by results and analysis in Section 6. Discussions and conclusion based on these results conclude the article in Section 7.

2. Research questions

This study aims to examine language and cultural aspects of Primary 4 national exams in relation to epistemic inclusion in the subjects of English language and natural science in Seychelles. The research was guided by the following questions:

i. How do the linguistic demands of the **question type** (multiple choice, short answers, free text etc) affect performance (Quantitative analysis)?
ii. How do **language structural aspects** such as sentence complexity and choice of vocabulary in the question and/or task instruction affect performance (Qualitative analysis)?
iii. To what extent, and how, do contextual reference points of the questions (images, types of examples, cultural setting etc.) take **cultural inclusivity** into account (or not)?
iv. What type of language approach (**structural or communicative**) can be understood from the marking schemes (particularly in the English exams)? Do the marking schemes acknowledge potential language challenges that being examined in an L2 involve?

3. Background – an overview of the language-in-education policies

The Seychelles education system is highly centralized, and national assessments are used to monitor progress and inform placement decisions. Major national exams are held every two to three years at the end of each key stage.[1] Exams are set and marked under the Ministry of Education's supervision and affect aspects such as streaming and selection for post-secondary schooling. The language-in-education policy formally promotes trilingualism, with Kreol Seselwa used as the main MoIA during the early years of primary education, and with English and French being part of the curriculum as subjects from Primary 1. In Primary 3, English takes over as MoIA but note that a policy change from

[1] There are 5 key stages in the Seychelles compulsory school system: Key Stage 1 (Crèche-Primary 2), Key Stage 2 (Primary 3-4), Key Stage 3 (Primary 5-6), Key Stage 4 (Secondary 1-3), Key Stage 5 (Secondary 4-5).

2017[2] provides for the use of English as MoIA from Primary 1 in the subject of mathematics if the school so chooses.

Although the Seychelles National Curriculum Framework (Ministry of Education, 2013) promotes inclusivity, equity and the recognition of Kreol Seselwa as central to learning and identity formation, there is a clear gap between these intentions and classroom realities, where a predominately English-only orientation increasingly dominates. While Kreol Seselwa is officially positioned as the main medium of instruction in the early years, its role is largely transitional and aimed at facilitating the acquisition of English and French rather than supporting long-term multilingual learning (Zelime and Deutschmann, 2016; 2025). Furthermore, issues related to English MoIA and inclusivity are not addressed at all in the Inclusive Education Policy (Ministry of Education, 2015). This document only discusses language in relation to inclusivity with reference to reading and writing difficulties and does not discuss or acknowledge limited proficiency in English as a potential exclusion factor.

Arguably then, the challenges learners face when English becomes the medium of instruction from Primary 3 are insufficiently addressed, particularly regarding the subject-specific language demands required for learning content subjects through English. In practice, there is little structured support for second-language learners, limited collaboration between language and subject teachers, and few classroom or assessment resources to support multilingual pedagogies. At the same time, official policies and reports suggest that the use of Kreol Seselwa in classrooms is actively discouraged (Deutschmann and Zelime, 2021, p.68). The English curriculum itself further reflects this tension, as it focuses primarily on structural and grammatical accuracy and the production of 'correct English', rather than on communicative competence or the language skills learners need to participate meaningfully in academic subjects (Zelime, 2022, p.20). Overall, this illustrates a significant disconnect between the multilingual and inclusive ideals expressed in the overarching national curriculum framework and the English-dominant practices that characterize everyday schooling.

4. Theory and previous research

Firstly, the theoretical framing of this study, namely epistemic inclusion, as defined by Barrett et al. (2025), refers to learners' participation in meaning-making within formal education. It has two interconnected dimensions: recognizing and valuing learners' own knowledge and that of their communities (Barrett, Milligan et al., 2025; Heggen et al., 2022), and ensuring access to, and engagement with, the formal curriculum. Grounded in sociocultural theories of learning (Daniels, 2001; Vygotsky, 1978), it emphasizes that new knowledge is built on learners' existing understandings and expands learning beyond

[2] see https://www.nation.sc/archive/252671/cabinet-decisions

content mastery to bridging different knowledge systems. This framing highlights the importance of engaging learners' language and meaning-making practices. The framework also assumes ensuring that learners are given all means possible to demonstrate their knowledge when being assessed. Here inclusive design of examinations is of primary importance.

Research focusing specifically on language issues and examinations in Sub-Saharan African (SSA) schooling systems remains relatively limited. Nevertheless, some studies have examined the challenges associated with the use of a second language in high-stakes assessments and the linguistic complexity of examination papers. In their analysis of eight, 2010 Certificate of Secondary Education Examination subject exams in Tanzania, for instance, Bartlett and Vavrus (2013) argue that the national examination system operates as a form of linguistic gatekeeping. They identify several problematic features related to exam language, including the frequent use of complex and elevated vocabulary such as *incidence*, *prevail*, *eradicate*, *incentive*, *initiated*, and *inevitable*, which may hinder learners' understanding of what questions require. The authors also note that examination texts commonly employ genre-specific discourse features, including passive constructions and dense nominalization, demanding levels of linguistic competence that extend beyond everyday conversational English. In addition, the exams rely heavily on decontextualized language and provide few supporting meaning-making resources, such as images or visual cues, which may further disadvantage second-language learners.

An additional issue identified by Bartlett and Vavrus (2013) concerns the use of unclear or unnecessarily complex instructions, which may negatively affect learners' performance across an entire examination. They illustrate this with an example from a bible studies examination: 'This paper consists of alternatives A and B. Each alternative comprises eight (8) questions. Confine yourself to only one alternative' (2013, p.106). The use of relatively complex lexical items such as *confine* and *comprises* makes instructions such as these inaccessible to less proficient learners. Furthermore, in some cases, poorly formulated questions also made it difficult for learners to determine precisely what was being asked. Ultimately, Bartlett and Vavrus (2013) argue that the academic English used in examinations effectively transforms subject-based assessments into language tests, thereby disadvantaging learners with limited exposure to formal English in classroom settings.

Similar conclusions are reached by Prophet and Badede (2009) in their experimental study of Grade 10 learners in Botswana, where English is a second language for almost all students. The researchers compared student performance of groups taking the original exams with groups taking modified versions with the same content but where language had been adapted for second language users. Their findings showed that improved readability generally enhanced student performance, with many revised questions producing increases of more than 10% in correct responses. However, performance actually declined on some of the modified items, leading the authors to conclude that although improved readability can benefit second-language learners, simplification must be approached carefully.

Rea-Dickins and colleagues (Rea-Dickins et al., 2013; Rea-Dickins and Yu, 2013) similarly explored the relationship between English proficiency and examination performance in Zanzibar through interview-based research focusing on the Form 2 National Assessment. At the time of the study, Kiswahili remained the language of learning and teaching throughout primary school, with a transition to English occurring at secondary level. Learners were asked to answer mathematics and science examination questions before discussing the questions in interviews designed to identify linguistic difficulties. The researchers subsequently modified the most challenging items and asked the same students to attempt the revised versions. Since interviews were conducted largely in Kiswahili, the researchers were also able to assess learners' subject knowledge independently of their English proficiency. The findings demonstrated that language-related difficulties in both interpreting questions and formulating responses in English often obscured learners' actual mathematics and science knowledge. Modifying the linguistic complexity of the questions positively affected overall performance. The research team later developed Kiswahili versions of the examination papers and learners performed better on these.

A comparable intervention is described by Heugh et al. (2017), who designed and trialled a multilingual assessment model for Grade 8 learners in South Africa. Because of prevailing education policy at the time, subject-content assessment could only be conducted in Afrikaans and English but the researchers nevertheless simplified the English used in the subject sections, and incorporated translations and glossaries in IsiXhosa. A follow-up questionnaire revealed that 56% of learners used the translations, and 85% of these learners considered them useful. Interestingly, some Afrikaans- and English-speaking learners also used the IsiXhosa translations, with 80% reporting that they found them beneficial. The study demonstrates that carefully designed multilingual assessments can support learners across a range of linguistic backgrounds and proficiency levels.

Two further studies examining language and assessment in content subjects are Zelime et al. (2018) and Zelime and Deutschmann (2019), both situated in the Seychelles context. These studies investigated how the language of assessment affects Primary 6 learners' ability to express subject knowledge in English and Kreol Seselwa, particularly in social studies, a discipline strongly connected to local culture and lived experience. Zelime et al. (2018) examined the impact of assessment language using a controlled within-group design in which learners completed the same Social Studies test in both English and Kreol Seselwa. The findings revealed an average performance increase of approximately 13% when learners answered in Kreol Seselwa, suggesting that they were significantly better able to communicate subject knowledge in their first language. Notably, this improvement occurred across all ability levels.

In a more qualitative study, Zelime and Deutschmann (2019) used a Systemic Functional Linguistics (SFL) analytical framework to analyse 308 written texts produced by Primary 6 learners in Kreol Seselwa and English. Their findings indicate that learners produced longer, richer, and more personally engaged texts in Kreol Seselwa, using broader locally-grounded vocabulary and more frequent first-person pronouns, suggesting stronger

identification with the content. By contrast, English-language texts were generally shorter, involved more code-switching, and reflected greater linguistic distance from the topic. Weaker learners, in particular, struggled to express culturally specific knowledge in English. The study therefore illustrates how mother-tongue writing enables learners to draw on personal experience, local knowledge, and emotional engagement.

5. Methods

Materials

The present study investigates national examination papers in the subjects of English and science from the years 2022-2024. In both subject areas, the examination papers were consistent in format and question types across the three sampled years. The examinations are described in more detail below.

In English, the reading paper (50%) and the writing paper (30%) were analysed. The listening paper, which constituted 20% of the final grade, was not included in the analysis. The reading paper consisted of eight exercises divided into approximately 50 sub-questions of varying complexity. These primarily included multiple-choice questions, such as matching words with images or short descriptive statements, fill-in-the-blank questions requiring learners to select the correct word from a set of alternatives, true/false questions, and multiple-choice reading comprehension questions based on short passages. These question types accounted for approximately 70% of the total marks. The remaining 30% consisted of short-answer questions based on text comprehension. The paper carried a total of 50 marks.

The writing paper consisted of two sections: Writing One (15 marks) and Writing Two (15 marks). In all examinations analysed, Writing One involved a postcard-writing task, while Writing Two required learners to produce a composition of approximately 80-100 words. Learners could choose from four topics with titles such as 'The most amazing adventure' and 'The person I admire the most'. Some options were also based on picture sequences. The duration of the paper was one hour and thirty minutes.

In science, the Key Stage 2 P4 General Science National Examination papers were analysed. These examinations consisted of two sections: Section A (20 marks), comprising 20 multiple-choice questions; and Section B (30 marks), comprising six structured questions. These longer questions contained several parts and included a range of question types, such as multiple-choice questions, true/false questions, single-word response questions, and short-answer free-text questions. The examinations lasted one hour and carried a total of 50 marks.

In addition to the examination papers, relevant examination reports were also analysed. These included the Key Stage 2 Examination Reports in English (Seychelles, 2022, 2023,

2024) and the Key Stage 2 Examination Reports in science (Seychelles, 2022, 2024).[3] Access was also obtained to all marking schemes associated with the examinations. Discussions of marking schemes, including what constituted acceptable and unacceptable answers, were incorporated into the examination reports and formed part of the analysis. The material included in the study is summarized in Table 1 below.

Table 1. Documents analysed

Context	Transition	Subjects Examined	Years	Exams	Other Documents
Seychelles	Transition in Primary 3 (age 8-9yrs)	English & science	2022 2023 2024	National Exams – Primary 4 (end of Key Stage 2) English (writing and reading only) Science (general)	(a) Examination reports (2022 and 2024 only for science) (b) Marking schemes

Method descriptions

Firstly, with reference to Research Question 1, information was retrieved from the examination reports regarding the percentage of learners who successfully answered each question item. By matching this data against the content of the examinations, each item could be classified according to question type, thereby providing an accurate overview of success rates across different question types.

To address Research Question 2, a qualitative analysis was conducted of questions that stood out in terms of success rates. Examples of both positive and negative outliers were selected for more detailed qualitative analysis, with particular focus on language complexity. Five such representative and illustrative examples from the different exams were selected, analysed for language complexity and are discussed below.

Focusing on Research Question 3, a multimodal analysis of images and texts was conducted to assess the potential familiarity of the cultural contexts represented. For example, consideration was given to whether the animals and plants depicted in the science exams were organisms with which learners were likely to be familiar. In the results, I give a quantitative overview of this analysis and illustrate tendencies using examples from both subjects.

Finally, with reference to Research Question 4, a close reading analysis was conducted of the various examination reports and marking schemes to identify instances where language aspects were discussed in the evaluation summaries and recommendations. In the English reports, particular attention was paid to the extent to which structural aspects such as

[3] The 2023 examination report in science was not availaible

grammatical correctness, spelling, punctuation, layout etc. rather than communicative aspects formed the basis for recommended marking practices. The overall aim was to determine the language orientation underpinning the subject. In the science reports, analysis focused primarily on whether/how language aspects were recognized as a potential challenge for learners.

6. Results

i) Question type analysis

Across the exam papers in both subject areas, a clear overall pattern emerges: the type of exam question strongly predicts learner performance, regardless of context or subject area. Closed-response question types, such as multiple-choice, true/false, and fill-in-the-blank (cloze test) questions, consistently show higher success rates than free text answers. The former question types require no writing and rely on comprehension and recognition rather than linguistic production. They thereby place lower linguistic demands on learners. Also note that multiple-choice questions with illustrations have a significantly higher pass rate than those without.

In contrast, questions requiring free text production, including one-word answers, short-answer sentences and longer structured written responses, such as shorter essays, show substantially lower success rates across all datasets. This pattern appears in both subjects investigated. Noteworthy here is that free-text questions, especially in the subject of English, constitute a relatively large proportion of the total exam mark, thereby lowering learners' overall scores and contributing to high failure rates.

Even within multiple-part questions, such as reading comprehension questions in English, where learners perform relatively well on the closed-format items, the shift to open-ended writing rapidly reduces success levels. Examination reports frequently attribute these difficulties to challenges in sentence construction, limited vocabulary, and insufficient mastery of academic writing.

Overall, the data suggest that exam performance is largely shaped by the linguistic demands of the question type. When writing demands increase, performance drops sharply. In contrast, when recognition-based or highly scaffolded formats are used, performance improves substantially. This pattern holds consistently across all years studied in both subjects. The specific data on which the above is based is summarized in Table 2. below. Note that the numbers in brackets represent the percentages of the total marks allocated to different question types.

Table 2: Success rates depending on question types in the different contexts

Question Type	Seychelles – P4	
	English	Science
Multiple choice – with figures/images	80% (12.5%)*	47% (30%)
Multiple choice – text only	50% (14%)	31% (14%)
True-false – with figures/images	78% (7%)	–
True-false – text only	66% (5%)	–
Match-type questions	–	42% (13%)
Cloze tests	50% (10%)	–
Free text – one word	–	38% (15%)
Free text – sentence	39% (15%)	31% (28%)
Free text – paragraph/essay	41% (36.5%)	–

*Numbers in brackets indicate total proportion of the exam marks that are made up of the specific question type

ii) Qualitative analysis of outlying questions

This section presents a more detailed analysis of five different types of questions to illustrate how different language aspects of questions might affect the learners' capacity to understand and answer questions correctly. The examples chosen were outliers that were deemed to be particularly linguistically challenging or accessible, i.e. questions where learners are likely to have performed particularly badly or well due to language issues.

Example 1 – Multiple-choice question, low success rate (Seychelles Science, 2022)

The example below (see Figure 1) was taken from the 2022 Seychelles science exam. Only 2% of the learners who answered this question managed to answer it correctly. The task constitutes a text-only multiple-choice question in the first section of the exam. Alternative B in Figure 1 below was the correct answer.

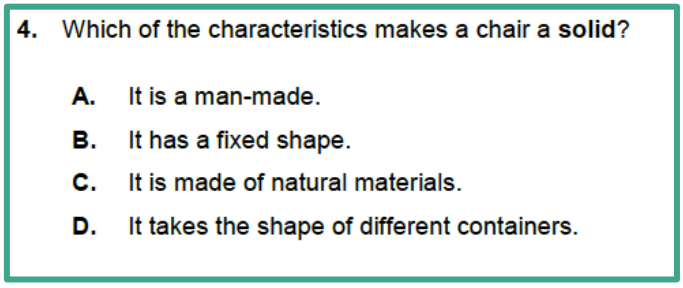


Figure 1. Negative outlier multiple choice science question 2022

The first issue that stands out with this question is the choice of vocabulary. The question contains many complex academic words such as *characteristics*, *containers*, *fixed shape*, and *materials* that many young learners may not fully understand. Secondly, the way the question is formulated, 'makes a chair solid' is cognitively more demanding than simply asking 'which characteristic is true of a solid?'. Furthermore, some of the answer alternatives, such as D – 'takes the shape of different containers', involves a complex passive syntactic structure and requires understanding both 'takes the shape' and 'containers', which are cognitively demanding phrases for young ESL learners. The correct answer – 'It has a fixed shape' – involves the use of abstract academic language and most young learners probably think of solids as keeping their shape or being hard. Overall, it is likely that language aspects, rather than the lack of understanding of what a solid is, has led to the poor results in this question. Also note that alternative A is grammatically incorrect and may not make sense because of this. To summarize, the biggest challenge here is arguably not the science but rather the academic vocabulary and filtering relevant from irrelevant information in complex sentences.

Example 2. Multiple choice, low success rate (Science, 2024)

The example in Figure 2 below was taken from the 2024 Seychelles science exam. The task is a text-only, multiple-choice question in the first section of the 2024 science exam with four alternative answers. Only a minority (about 30%) of the learners answered this question correctly. Alternative A in Figure 2 below was the correct answer.

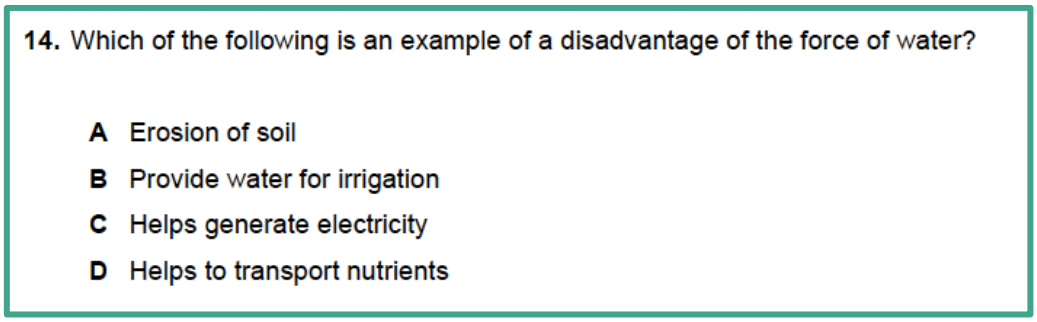


Figure 2. Negative outlier multiple choice science question 2024

Arguably this is another example of where language is the main hurdle leading to failure rather than a lack of understanding of what the force of water can accomplish. The choice of words such as *disadvantage*, *erosion*, *provide*, *irrigation*, *generate*, *transport*, and *nutrients* does not take the proficiency level of the average learner in Primary 4 into account. When checking the complexity of this question in a CEFR level analyser,[4] we get an average level of B2 competence, i.e. upper intermediate proficiency, a level of English expected from advanced learners (typically 15+), and not children of 8-9 years of age who have been taught in English for two years.

[4] See https://www.langpodtools.com/tools/cefr-analyzer

Example 3. Multiple choice, high success rate (Science, 2024)

In contrast to the above questions, the multiple-choice question from the 2024 science exam (illustrated in Figure 3 below) was relatively successful; 65% of learners managed to answer this question correctly. The task is concise, contains clear instructions and the use of images help the learners overcome potential language difficulties associated with scientific words such as *organs*.



Figure 3. Positive outlier multiple choice science question 2024

Example 4. Short-answer reasoning, low success rate (Question 21c) ii, Science, 2024)

The task in Figure 4 below was only answered correctly by 17% of learners taking the exam. Arguably, the problem here is not that learners do not know that salt and sugar dissolve in water and that lentils do not, but rather that learners have problems formulating the reasoning as to why Paul's statement is incorrect. I would argue that a question simply asking which of the three substances did not dissolve in water would have rendered a much higher success rate.

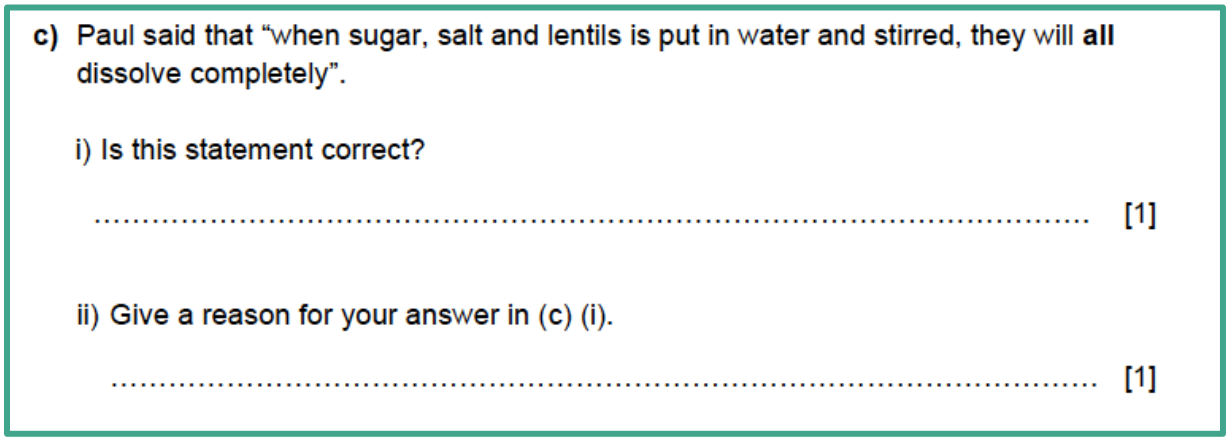


Figure 4. Negative outlier short answer science question 2024

Example 5. Short-answer question, low success rate (English reading, 2024)

The next example is taken from a short-answer comprehension question in the English reading paper 2024. The success rate here was a mere 7%. The text which learners had to read was about Lionel Messi and comprised about 500 words. The first three paragraphs of this text (about 200 words) list several impressive accomplishments of the young Messi's football career (5-17 years). The item chosen for analysis here (item 7 of Question 8B) asks: *In the history of Spanish football as a young player, what did he accomplish?*

According to the marking scheme, the answer had to include a phrase to the effect that he was the youngest player ever to score for Barcelona. There are problems with this question. Firstly, the choice of the word 'accomplish' is questionable, especially given that the examiners are after some kind of record. Furthermore, the structure of the question is highly confusing. The phrase 'in the history of Spanish football as a young player' mixes two ideas, historical context and his early career, without clearly signalling what it refers to. Furthermore, it is unclear what aspect of the young player's accomplishments is the main focus. Arguably, the use of 'history' here could infer a record of some sort, but this is not made explicit. Given the fact that the comprehension reading text lists at least six accomplishments, it is indeed challenging for young learners to pick the exact 'correct' information from a long and complex text, given such vague and overly complex instructions.

In summary, the analysis of sentence structure and choice of vocabulary in the questions reveals a lack of adaptation for the language proficiency levels in English expected from young L2 learners who have only been taught in English for two years. Furthermore, there were many examples of grammatical inconsistencies in the exam questions further adding to potential confusion. In addition, some answers which were deemed to be correct according to the marking schemes were in fact incorrect.[5]

iii) Cultural inclusivity

Below, I will discuss examples of how the cultural context of the questions potentially have facilitated/hindered understanding. Overall, the following tendencies emerged.

In the English papers, 39 % of the questions referred to foreign cultural references. This category constituted questions that included text or image references which connected to contexts outside Seychelles. Figure 5 below, for example, was the reference image for several true and false questions in the 2024 English reading paper. There are no zoos in Seychelles. The image also contains a cityscape totally absent in Seychelles and all the people in the image are white.

[5] For example, see Question 6 in the 2022 science exam where amphibians are described as having 'moist scaly skin'.

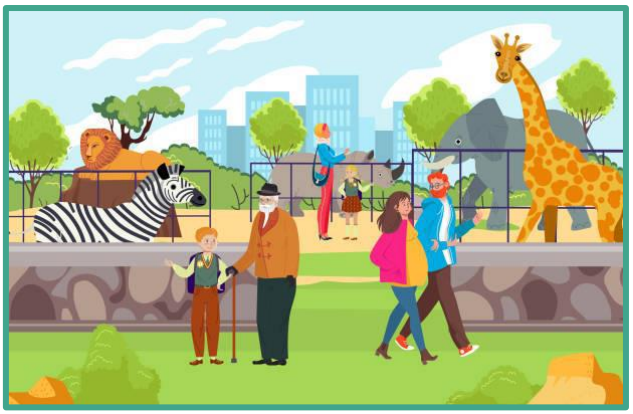


Figure 5. Question reference in the 2024 reading paper

Local cultural references were present in 27% of the English paper questions. One example was a writing task where learners had to write a postcard describing a trip to a local island – La Digue. Another example was a reading comprehension task based on a local advert for a National Day Bazaar organised by the Seychelles Trading Company (STC). Children will be familiar with this type of event, and images, logotypes as well as activities would be recognizable to them.



Figure 6. A local advertisement used as topic for English reading comprehension exercise in 2023 exams

Finally, in the investigated English papers, 34% of the questions had neutral reference points where no specific cultural context could be identified.

In the science papers, the bulk of the questions, 72%, were culturally neutral, whereas 17% of the questions referred to foreign contexts. The question in Figure 7 below from the 2022 science exam for example, required learners to analyse a food web from what probably is a temperate climate region. There are no deer, squirrels, foxes, jaguars, pine trees or berries in Seychelles, and it may thus be difficult for learners to relate this web to their outside school knowledge.

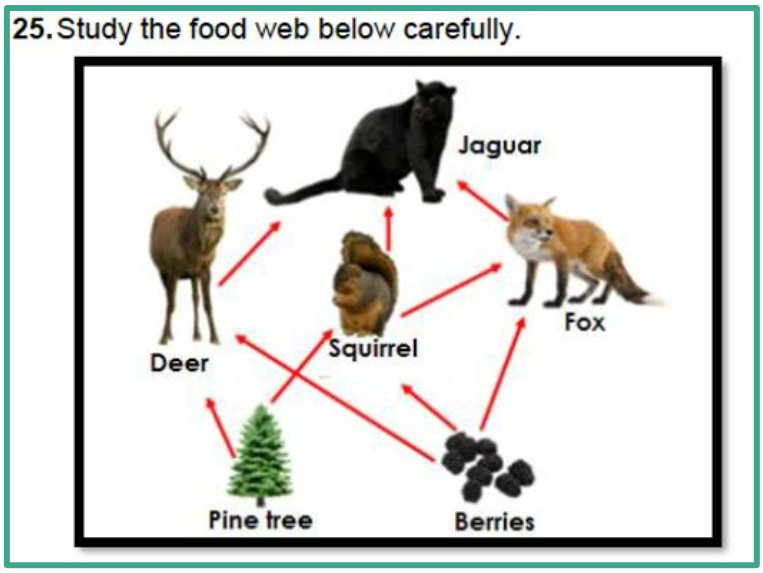


Figure 7. Foreign context reference in the 2022 science paper

There were several examples of references to foreign animals and plants in the exam, examples which easily could have been replaced by more local examples.

Questions with local references made up 11% of the exam questions in the investigated science exams. Most of these examples constituted references to local animals and plants. In the example below (Figure 8) a local plant is used successfully to guide learners in the right direction in a multiple-choice question from the 2024 science exam. The question had a success rate of 42%, which was surprisingly high given the complexity of the concept tested, i.e. 'characteristics of living things'. All the concepts listed as answer alternatives are complex and abstract, both linguistically and conceptually – *reproduction*, *sensitivity*, *growth* and *respiration*. The use of the term *characteristics* adds to the level of abstraction and is arguably beyond the proficiency level of Primary 4 learners with limited command of English. However, the inclusion of the illustrated example, the mimosa plant also named in brackets in Kreol Seselwa (i.e. *sansib*) is arguably good practice and an example of how local contextual knowledge can be used to illustrate complex scientific concepts. The mimosa plant is familiar to most Seychellois children. When you touch the plant, it folds its leaves, and children know this as they have experienced it themselves. Thereby they can relatively easily relate this to the concept of touch, i.e. sensitivity. This is further helped by the fact that *sansib* means 'sensitive'.

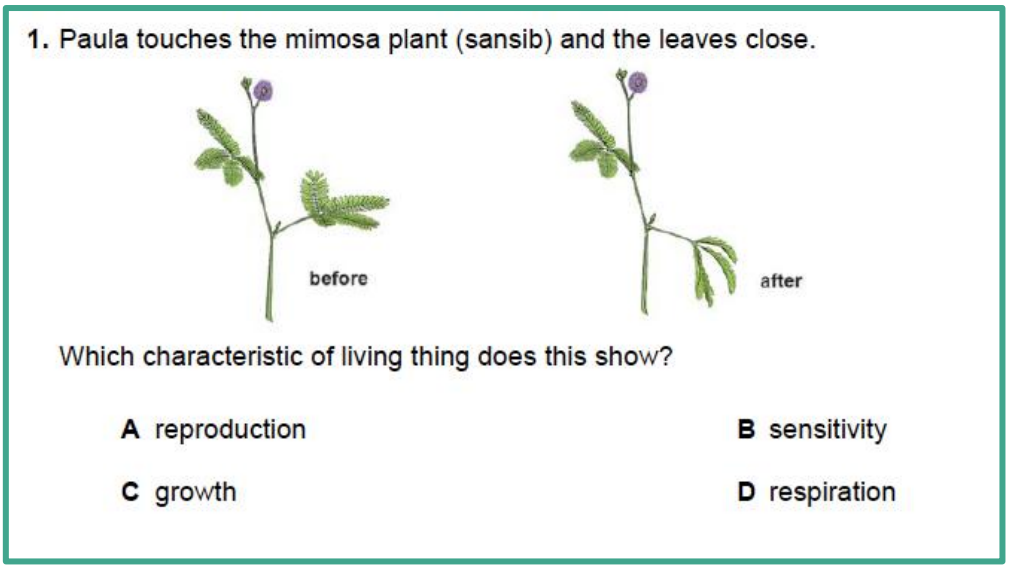


Figure 8. An example from the 2024 science paper of good use of illustrations and local context to help learners

iv) Analysis of marking schemes – structural or communicative language approaches?

In this part, marking schemes and performance reports are analysed, looking for indications of whether the examiners' orientation to language was structural or communicative.

Across the English reports, a discourse emerges which emphasizes a structural approach where language is treated primarily as a system of rules, formats, and correct forms to be mastered. For example, 'skills in grammatical structures, proper spelling, punctuation' are emphasized as well as 'appropriate layout' (English Assessment Report, 2022). Similarly in the 2023 report the main focus on writing tasks is described as a means for candidates to demonstrate 'their proficiency in grammatical structures, accurate spelling, proper punctuation, appropriate style and register, as well as using a suitable layout' (2023, p.16). The emphasis on form and structure is further illustrated in the marking schemes, where in the postcard task, for example, nine out of fifteen marks were awarded solely for features such as layout, grammar, and punctuation. Such marking criteria underscore that form and accuracy are prioritized in the system. The examination reports also reveal that children experience these aspects as challenging: 'The candidates did not perform as expected in postcard writing. It seemed that the layout was not well mastered by the candidates. As for writing two the main issues was with the vocabulary and tenses to be used.' (English Assessment Report, 2022, p.20). This interpretation is further supported by comments regarding weaknesses in learners' performance. 'Issues with tense usage' are emphasized, 'with candidates often using incorrect verb forms' motivating 'further reinforcement of grammatical concepts, particularly with regard to tense agreement' (English Assessment Report, 2023, p.20).

Having said this, there are, however, competing discourses in the exam reports. These are particularly evident in the 'Suggestions for Improvement' sections of the reports, where more communicative approaches to language can be discerned. In the 2022 report, for example, teachers are urged to '[m]ake provisions during the lessons for pupils to re-count and re-tell which will develop their writing skills' (p.21). This focus on oral communicative skills to communicate a story or an event clearly stands out. In the same report, teachers are also encouraged to '[e]nsure that pupils are exposed with opportunities to stimulate their thinking skills' (p.21). In the 2024 report teachers are urged to '[e]ncourage creative and personal expression: Provide engaging writing prompts that allow pupils to write about familiar experiences or imaginative scenarios' (p.29). This last suggestion connects to a communicative approach where local contexts and experiences are given emphasis.

While the science assessments reports recognize problems in performance – 'the overall performance for 2022 shows that the mastery of scientific knowledge, understanding and reasoning are regressing remarkably' (Science Assessment Report, 2022, p.12), and '[t]he number of blank and incorrect responses in Section B highlight that higher-order thinking and interpretation skills are not yet adequately developed across the cohort' (Science Assessment Report, 2024, p.11), there is no specific recognition that it could be due to language issues such as insufficient proficiency in English to communicate scientific ideas and the use of inaccessible language in the question constructions. Given that the 2024 report lists 'communication of ideas, information and the use of the scientific language' (p.2.) as one of three main competencies assessed in the paper, it is remarkable that none of the recommendations for improvements listed in the last section of this report address language issues related to English MoIA.

Taken together, the reports reveal a lack of recognition of the challenges involved in L2 MoIA, as well as a consistent structural orientation to language whereby correctness, rule-governed forms, fixed formats, and monolingual English use are foregrounded, while communicative meaning-making, flexibility, and multilingual resources receive little or no recognition. Consequently, I would argue that English proficiency functions as a pervasive gatekeeper throughout the assessment system.

7. Discussion and conclusion

The findings of this study clearly show that linguistic complexity may play a decisive role in shaping learners' performance in exams. Across all papers investigated, a consistent pattern emerges whereby closed-response question types, such as multiple choice, true/false, and fill-in-the-blank, lead to substantially higher success rates than free-text questions, which display very low success rates. Questions which demand complex grammar manipulation seem to be particularly problematic. This supports earlier work arguing that when examination questions prioritize language form over conceptual

understanding, they effectively transform subject tests into assessments of L2 proficiency (cf. Bartlett and Vavrus, 2013; Prophet and Badede, 2009).

Many of the items associated with particularly poor performance in this study contained dense academic vocabulary, abstract syntax, and/or cognitively demanding discourse structures, making them inaccessible to learners with limited exposure to English as the language of learning and testing. These patterns are mirrored in studies of various other SSA exam systems and not unique to Seychelles (cf. Prophet and Badede, 2009; Bartlett and Vavrus, 2013; Rea-Dickins and Yu, 2013). Here, it is tempting to speculate that these linguistic patterns are a result of exams being largely based on formats based on exams written for native English speaking contexts, but further research is needed to confirm this.

Arguably, the direct cultural connection to former colonial contexts is however evident in the use of images and other cultural references. The food web in Figure 6, for example, is probably a direct copy from a Western source. This is understandable given that many learning materials are produced abroad and teachers use examples from these in the exams.

There are, however, several encouraging findings in this study. The data show that some questions are (to some extent), adapted to L2 learners. Here, questions supported by illustrations and familiar local contexts help learners to better be able to demonstrate their knowledge. Similarly, questions using clear instructions, simple syntax, and less complicated vocabulary seem to lead to better success rates. These findings echo results from multilingual assessment interventions in Botswana, Zanzibar, and South Africa (cf. Prophet and Badede, 2009; Rea-Dickins and Yu, 2013; Heugh et al., 2017), and point to the benefits of adapting question format and language to better suit the levels of proficiency of the learners.

Overall, however, it is surprising that not more attention is paid to examination design which accommodates the needs of L2 learners. As is the case in the overarching curriculum framework and subject curricula (see Deutschmann and Zelime, 2021), specific issues related to L2 MoIA challenges and the securing of epistemic inclusion seem to be a blind spot in the system and linguistic gatekeeping remains a major problem.

Recommendations

Based on the study's findings, several recommendations emerge for improving the fairness and validity of high-stakes examinations in the Seychelles contexts. These are listed below:

1. *Reduction of unnecessary linguistic load:* Exam developers should systematically avoid the use of low-frequency academic vocabulary and instead opt for clearer high-utility words. They could also opt for more straight forward syntax avoiding complex embeddings, passive constructions and nominalizations.

2. *Questions should be designed to test what they set out to test*: In particular, in the science subjects, it is important to ensure that each question primarily tests content

knowledge rather than English skills. Similarly, in English exams questions testing communicative skills should focus on meaning rather than form.

3. *Marking schemes should adopt a more communicative orientation***:** We recommend that exam developers shift emphasis away from rigid structural accuracy and monolingual frameworks. Conceptually correct answers should be accepted even when phrased in non-standard English or when translanguaging strategies are used.

4. *Integrate contextual and visual scaffolding*: Images, familiar local examples, and/or bilingual glossaries can be used to support comprehension.

5. *Piloting and post-analysis*: Regular piloting of exam items with representative learner groups should be institutionalized so that problematic items can be identified and revised before high-stakes administration. Similarly, the success rates of different questions should be analysed with regard to language aspects after exams to identify problematic issues.

Finally, a more radical approach would be for policymakers to consider the strategic introduction of bilingual assessment components. Evidence from Seychelles and South Africa shows that such measures significantly enhance learners' ability to demonstrate subject understanding. Language support during exams should also be available in all subjects other than language subjects. Taken together, these recommendations aim to mitigate the linguistically exclusionary effects identified in this study. Ultimately, addressing linguistic gatekeeping in exam design is not only a matter of assessment quality but a question of epistemic equity, ensuring that high-stakes examinations measure what learners actually know.

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